Marine Engineering Knowledge (MEK)

176 Practice Questions and Answers for STCW Exams

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Practice Set 1

Q1. Boiler is related to ___________________

A) Steam Generation
B) Human waste
C) Storing Vegetables
D) Steering

Correct Answer: A (Steam Generation)

Explanation: A boiler is a crucial piece of equipment onboard a ship responsible for generating steam. This steam is utilized for various purposes, including propulsion in steam-powered vessels and for heating systems.

Q2. Which of the following is bottom most part of a diesel engine?

A) Connecting Rod
B) Piston
C) Turbo charger
D) Exhaust valve

Correct Answer: A (Connecting Rod)

Explanation: The connecting rod is positioned between the piston and the crankshaft. It is the bottom-most component among the choices provided that is directly involved in transmitting the linear motion of the piston to the rotating motion of the crankshaft.

Q3. Main Engine is part of the __________________ system

A) Ballast
B) Electrical
C) Propulsion
D) Bilge

Correct Answer: C (Propulsion)

Explanation: The main engine is the primary source of power for a ship's movement, providing the necessary force to turn the propeller. Therefore, it is an integral part of the propulsion system.

Q4. Domestic Fridge is related to ___________________.

A) Human waste
B) Steering
C) Starting of Main Engine
D) Storing Vegetables

Correct Answer: D (Storing Vegetables)

Explanation: A domestic refrigerator onboard is used for the primary purpose of storing perishable items like vegetables, ensuring *their freshness. It is a component of the ship's provisions and galley facilities.*

Q5. Which of the following is top most part of a diesel engine?

A) piston
B) cylinder head
C) crank shaft
D) crankcase door

Correct Answer: B (cylinder head)

Explanation: The cylinder head is the topmost component of a diesel engine cylinder, sealing the combustion chamber and housing the valves and injectors. It sits above the piston and cylinder liner.

Q6. Which of the following is lower most part of a diesel engine?

A) turbo charger
B) exhaust valve
C) bed plate
D) piston

Correct Answer: C (bed plate)

Explanation: The bed plate forms the base of the diesel engine structure, supporting the crankshaft and carrying the main engine load. It is the lowest structural part of the engine assembly.

Q7. Which of the following is within the engine room?

A) paint storage room
B) CO2 Room
C) cargo pump room
D) purifiers room

Correct Answer: D (purifiers room)

Explanation: The purifiers room (or centrifuge room) is dedicated to housing the machinery used for purifying fuel and lubricating oils, essential for the operation of the main engine and auxiliary machinery. This room is typically located within the engine room complex.

Q8. Which of the following is not within the engine room?

A) fresh water generator
B) fire pump
C) fridge room
D) refrigeration machinery

Correct Answer: C (fridge room)

Explanation: The fridge room, used for storing provisions, is typically located in the galley or on deck, not within the engine room where machinery operates. The fresh water generator, fire pump, and refrigeration machinery are all essential engine room equipment.

Q9. Which of the following is not within the engine room?

A) boiler
B) mooring winch
C) air compressor
D) generator engine

Correct Answer: B (mooring winch)

Explanation: The mooring winch is deck machinery used for handling mooring lines and anchors, and is therefore not located within the engine room. The boiler, air compressor, and generator engine are all key components found in the engine room.

Q10. Which of the following is not a job of an engine room rating?

A) assist in watchkeeping
B) assist in watchkeeping
C) assist in maintenance work
D) operate main engine

Correct Answer: D (operate main engine)

Explanation: Operating the main engine is a responsibility typically held by the engineering officer of the watch or a senior engineer, not an engine room rating. Ratings assist in watchkeeping duties, perform maintenance tasks, and execute orders from their superiors.

Q11. Which of the following is job of an engine room rating?

A) start and stop engine
B) overhaul air compressor
C) assist in maintenance work
D) report to bridge about engine condition

Correct Answer: C (assist in maintenance work)

Explanation: An engine room rating's primary duties include assisting in watchkeeping, performing routine maintenance under supervision, and carrying out specific tasks assigned by engineers. Starting and stopping the main engine, overhauling components, or reporting directly to the bridge are generally not their responsibilities.

Q12. Boiler is related to ___________________.

A) Storing Vegetables
B) Human waste
C) Steam Generation
D) Steering

Correct Answer: C (Steam Generation)

Explanation: A boiler's primary function is to generate steam, which can be used for various purposes on board, such as heating, driving machinery, or powering auxiliary equipment. It is not involved in storing vegetables, handling human waste, or steering the vessel.

Q13. Sewage Treatment Plant is related to ___________________.

A) Starting of Main Engine
B) Steering
C) Human waste
D) Storing Vegetables

Correct Answer: C (Human waste)

Explanation: The Sewage Treatment Plant (STP) is specifically designed to process human waste generated by the crew and passengers, treating it to meet environmental discharge standards as per MARPOL regulations. It is unrelated to engine starting, steering, or food storage.

Q14. What is not a system

A) Boiler
B) Purifier
C) Air compressor
D) Main Engine

Correct Answer: D (Main Engine)

Explanation: The main engine is a complete functional unit that propels the ship. The boiler, purifier, and air compressor are individual systems or machines that are often part of the ship's overall engineering setup, but the main engine itself is the primary propulsion machine.

Q15. A "BOLT" has

A) Hexagonal head and threads at other end
B) Thread at both ends
C) Slotted head and threads at other end
D) Double pitch threads

Correct Answer: A (Hexagonal head and threads at other end)

Explanation: A bolt is a fastener characterized by external screw threads and typically a hexagonal head, designed to be inserted through unthreaded holes in assembled parts and secured by a nut. It is not a threaded rod or a screw with a slotted head.

Q16. "Cuts" are caused by

A) Flat spanners
B) Blunt chisels
C) Hammers
D) Sharp objects

Correct Answer: D (Sharp objects)

Explanation: Cuts, abrasions, and lacerations are typically caused by sharp objects such as knives, burrs on metal, or broken glass. While hammers and blunt chisels can cause blunt force trauma, they are not the direct cause of cuts.

Q17. A DIE can be is used to make threads on a pipe

A) True
B) False

Correct Answer: A (True)

Explanation: A die is a cutting tool used to create external screw threads on pipes or rods. Therefore, it is indeed used to make threads on a pipe, typically for connections or fittings.

Q18. For lifting a heavy load, you will use

A) Eye bolt
B) Air Cylinder
C) Hydraulic Jack
D) Chain Block

Correct Answer: D (Chain Block)

Explanation: A chain block is a mechanical device used for lifting heavy loads, consisting of a chain and pulley system. While an eye bolt can be used for lifting, it's a component, and a hydraulic jack lifts by fluid pressure; a chain block is the general tool for heavy lifting.

Q19. A TAP is used to make internal threads inside a drilled hole

A) True
B) False

Correct Answer: A (True)

Explanation: A tap is a tool used to create or repair internal screw threads, typically by being inserted into a pre-drilled hole and rotated. It is the counterpart to a die, which creates external threads.

Q20. A "STUD" has

A) Threads at one end with a slot at the other end
B) Double pitch threads
C) Hexagonal head and thread at the other end
D) Thread at both ends

Correct Answer: D (Thread at both ends)

Explanation: A stud is a threaded fastener that has threads on both ends and no head, used for connecting two components where nuts are typically threaded onto both ends. It differs from a bolt, which has a head and threads at one end for a nut.

Practice Set 2

Q1. Which of this is not fitted on the cylinder head of a diesel engine

A) exhaust valve
B) turbo charger
C) starting valve
D) indicator valve

Correct Answer: B (turbo charger)

Explanation: A turbocharger is an exhaust gas-driven turbine that forces more air into the engine cylinders to increase power and efficiency. It is not a component directly fitted on the cylinder head itself, unlike the exhaust valve, starting valve, and indicator valve.

Q2. Which of this is not a moving part of a diesel engine

A) piston
B) crankshaft
C) crankcase
D) connecting rod

Correct Answer: C (crankcase)

Explanation: The crankcase is the housing for the crankshaft and part of the engine block; it is a stationary component. The piston, crankshaft, and connecting rod are all essential moving parts within the diesel engine's power cycle.

Q3. Generator engine on board ship is started using

A) hydraulic oil motors/pressure
B) compressed air
C) turbo generator
D) air compressor

Correct Answer: B (compressed air)

Explanation: Generator engines on ships are typically started using compressed air stored in receiver tanks, which engages the engine's starting mechanism. Hydraulic systems or air compressors are used to supply this air, but the immediate starting medium is

Q4. Purpose of fresh water circulation through diesel engine cylinder is to

A) prevent corrosion
B) remove heat of combustion
C) clean exhaust gases
D) lubricate moving part

Correct Answer: B (remove heat of combustion)

Explanation: The primary purpose of circulating fresh water through a diesel engine's cooling system is to absorb and remove the immense heat generated by combustion, thereby maintaining optimal operating temperatures. It does not primarily prevent corrosion or clean exhaust gases.

Q5. Turning gear on main engine is provided for

A) releasing parts if jammed
B) turning the engine during overhaul
C) remove exhaus gases from cylinder
D) ensure turning of propeller

Correct Answer: B (turning the engine during overhaul)

Explanation: The turning gear is provided on the main engine to allow slow rotation of the crankshaft during maintenance, overhaul, or pre-starting checks. This ensures even cooling, allows access to components, and prevents seizure.

Q6. Exhaust gas economiser produces

A) fresh water
B) steam
C) inert gas
D) electricity

Correct Answer: B (steam)

Explanation: An exhaust gas economizer is a heat exchanger that recovers waste heat from the exhaust gases of the main engine or auxiliary boilers to generate steam or preheat feedwater. It is not involved in producing fresh water, inert gas, or electricity directly.

Q7. Turbo chargers on diesel engine

A) acts as fuel charger
B) decreases the pressure of intake air
C) increases pressure of intake air
D) increases RPM

Correct Answer: C (increases pressure of intake air)

Explanation: Turbochargers are used to increase the power output of diesel engines by compressing and forcing more intake air into the cylinders. This results in higher volumetric efficiency and improved combustion, not a decrease in air pressure.

Q8. Exhaust gases from the main engine go to the atmosphere through

A) inert gasgenerator
B) oil fired boiler
C) exhaust fan
D) exhaust gas boiler

Correct Answer: D (exhaust gas boiler)

Explanation: Exhaust gases from the main engine, after potentially passing through an exhaust gas boiler (economizer) for heat *recovery, are released to the atmosphere through the ship's exhaust uptake or funnel. An inert gas generator or exhaust fan serves different purposes.*

Q9. Relief valves are provided on main engine cylinders for protection against high

A) pressure
B) exhaust temperature
C) speed
D) level of water

Correct Answer: A (pressure)

Explanation: Cylinder head relief valves are safety devices designed to protect the engine from potentially catastrophic over-pressurization within the combustion chamber. This can occur due to excessive fuel injection or other abnormal combustion events, not high exhaust temperature or speed.

Q10. Function of a cylinder head relief valve is to

A) prevent crankcase explosion
B) stop the engine
C) reduce the speed of engine
D) release excessive pressure

Correct Answer: D (release excessive pressure)

Explanation: A cylinder head relief valve is a safety device that opens automatically to release excessive pressure from within the cylinder during operation. This prevents damage to the engine components and potential hazards from over-pressurization.

Q11. Function of crankcase relief door is to

A) release pressure from crankcase
B) stop the engine
C) reduce the speed of engine
D) detect the mist inside the crankcase

Correct Answer: A (release pressure from crankcase)

Explanation: Crankcase relief doors are designed as safety vents to rapidly release any abnormal pressure build-up within the crankcase, which can be caused by a crankcase explosion. This prevents damage to the engine casing and potential injury.

Q12. If cooling water becomes less in a diesel engine the engine will

A) make more noise
B) cool down too much
C) get overheated
D) not start

Correct Answer: C (get overheated)

Explanation: Insufficient cooling water in a diesel engine leads to inadequate heat removal from the combustion process, causing the engine's temperature to rise beyond its safe operating limits, resulting in overheating. Proper cooling is essential for engine longevity and performance.

Q13. Diesel engine bearing will get damaged if supply of ___________ is not enough

A) air
B) fuel
C) water
D) lubricating oil

Correct Answer: D (lubricating oil)

Explanation: Bearings in a diesel engine rely on a continuous supply of lubricating oil to create a film that separates moving surfaces, reducing friction and wear. A lack of lubricating oil will lead to metal-to-metal contact, overheating, and eventual bearing seizure or damage.

Q14. Speed of diesel engine is adjusted by changing the supply of

A) air
B) fuel
C) water
D) lubricating oil

Correct Answer: B (fuel)

Explanation: The speed of a diesel engine is primarily regulated by controlling the amount of fuel injected into the cylinders. Increasing fuel supply increases power and speed, while decreasing it reduces them. Air supply is also important but fuel is the primary control.

Q15. While diesel engine is operating ,one of the important pressure to check is ______

A) lubricating oil
B) sarting air
C) exhaust gas
D) steam pressure

Correct Answer: A (lubricating oil)

Explanation: While checking various parameters is important, the lubricating oil pressure is a critical indicator of proper lubrication, essential for preventing engine damage. Low oil pressure signals a potential problem with the lubrication system and requires immediate attention.

Q16. One of the function of lubricating oil in a diesel engine is to reduce

A) exhaust temperatures
B) friction between moving parts
C) wexhaust gases
D) weight of the engine

Correct Answer: B (friction between moving parts)

Explanation: The primary function of lubricating oil in a diesel engine is to reduce friction between moving parts, thereby minimizing wear and heat generation. It also provides cooling and sealing, but friction reduction is its most crucial role.

Q17. Function of crankcase lubricating oil in a diesel engine is to ____

A) improve combustion
B) prevent corrosion
C) supply power
D) turn crankshaft

Correct Answer: B (prevent corrosion)

Explanation: Lubricating oil within the crankcase serves multiple purposes, including lubricating the crankshaft and connecting rod *bearings, cooling these components, and preventing rust and corrosion on internal surfaces. It does not directly improve combustion or supply power.*

Q18. The lower most ring on a diesel engine piston is called

A) split ring
B) compression ring
C) pressure ring
D) oil scraper ring

Correct Answer: D (oil scraper ring)

Explanation: The bottom-most ring on a piston, typically featuring a tapered or spring-loaded design, is the oil scraper ring. Its function is to scrape excess lubricating oil from the cylinder walls during the piston's downstroke, preventing it from entering the combustion chamber.

Q19. For good combustion,the color of the exhaust from diesel engine should be

A) light brown
B) colourless
C) light grey
D) bluish white

Correct Answer: B (colourless)

Explanation: For efficient combustion in a diesel engine, the exhaust gases should be colourless, indicating complete burning of the fuel. Any discolouration, such as black, blue, or white smoke, signifies incomplete combustion or other issues needing attention.

Q20. Ships propeller is fitted to

A) stern tube
B) intermediate shaft
C) tail shaft
D) crankshaft

Correct Answer: C (tail shaft)

Explanation: The ship's propeller, which provides propulsion by converting engine power into thrust, is directly attached to the end of the tail shaft. This shaft extends from the gearbox or engine through the stern tube to the propeller.

Q21. The type of engine normally used for generators on a ship is _____________

A) 4 stroke
B) 2 stroke
C) 6 stroke
D) medium stroke

Correct Answer: A (4 stroke)

Explanation: Four-stroke engines are generally preferred for ship generators due to their better fuel efficiency, smoother operation, and lower emissions compared to two-stroke engines, which are typically used for main propulsion in larger vessels.

Q22. When starting engine on diesel oil, before injection the oil is

A) heated to same temp s heavy oil
B) heated if necessary in cold climate
C) passed through the purifier
D) mixed with steam

Correct Answer: B (heated if necessary in cold climate)

Explanation: Diesel oil, used for generators and smaller engines, has a lower ignition point than heavy fuel oil and typically does not require significant heating for injection. It may require warming in very cold climates, but not to the extent of HFO, nor is it typically mixed with steam.

Q23. A cam shaft on diesel engine operates

A) fuel injection pump
B) flywheel
C) fuel valves
D) starting valves

Correct Answer: A (fuel injection pump)

Explanation: The camshaft, driven by the engine's timing gears or chain, has lobes that push open the valves. It directly operates the fuel injection pumps in many diesel engine designs, ensuring precise fuel delivery timing.

Q24. Thrust block for main engine is fitted at the

A) forward side of engine
B) aft side of engine
C) end of the cam shaft
D) tail shaft

Correct Answer: B (aft side of engine)

Explanation: The thrust block is a specialized bearing designed to absorb the significant axial thrust generated by the propeller and transmitted through the propeller shaft. It is fitted at the aft end of the engine, in line with the shafting, to transfer this force to the ship's structure.

Q25. Function of a cylinder head relief valve is to

A) Release excessive pressure
B) Reduce the speed of the engine
C) Stop the engine
D) Prevent crankcase explosion

Correct Answer: A (Release excessive pressure)

Explanation: A cylinder head relief valve is a safety device designed to automatically open and release excessive pressure that may build up inside the engine cylinder during operation. This prevents damage to the cylinder head and liner and potential engine failure.

Q26. The purpose of Injector valve on a generator engine is for _____

A) Safety
B) Fresh air intake
C) Fuel inlet
D) Compress air

Correct Answer: C (Fuel inlet)

Explanation: The injector valve, also known as a fuel injector, is responsible for injecting a precisely metered amount of fuel into the engine cylinder at the correct time and pressure for combustion. Its primary role is fuel delivery, not safety or air intake.

Q27. Exhaust valve in a Diesel engine ____

A) Stays in place
B) Opens and closes
C) Rotates in place
D) Moves up and down

Correct Answer: B (Opens and closes)

Explanation: The exhaust valve in a diesel engine is a critical component that opens to allow burned exhaust gases to exit the cylinder and closes to seal the combustion chamber during the compression and power strokes. Thus, it must open and close cyclically.

Practice Set 3

Q1. Thrust block for main engine is fitted at the

A) End of the cam shaft
B) Tail Shaft
C) Forward side of Engine
D) Aft side of Engine

Correct Answer: D (Aft side of Engine)

Explanation: The thrust block is a substantial bearing that absorbs the significant axial thrust generated by the propeller and transmitted through the tailshaft. It is strategically located on the aft side of the engine, often integrated with the gearbox or stern tube, to efficiently transfer this force to the ship's structure.

Q2. Piston in a Diesel engine ________

A) Stays in place
B) Moves up and down
C) Oscillates
D) Rotates in place

Correct Answer: B (Moves up and down)

Explanation: In a diesel engine's operational cycle, the piston is designed to reciprocate, meaning it moves linearly up and down within the cylinder. This reciprocating motion is fundamental to converting the energy from fuel combustion into rotational mechanical work via the crankshaft.

Q3. The purpose of Exhaust Gas in a Diesel engine is to ___

A) Reduces friction
B) Increases air supply
C) Turns turbo charger
D) Gives power

Correct Answer: C (Turns turbo charger)

Explanation: The primary purpose of the exhaust gases, which are at high temperature and pressure, is to drive the turbine wheel of the turbocharger. This rotation then powers the compressor, which forces more air into the cylinders, thereby increasing engine power and efficiency.

Q4. What is to be checked on a running Main engine , during watch.

A) Exhaust temperature
B) Oil coming out of water side
C) Steering angle
D) Discharge pressure of liquid

Correct Answer: A (Exhaust temperature)

Explanation: During a watch, monitoring the exhaust temperature of the main engine is crucial. Abnormal fluctuations can indicate potential issues like cylinder imbalance, cooling problems, or combustion irregularities, allowing for early intervention.

Q5. Air starting valve in a Diesel engine _____

A) Oscillates
B) Moves up and down
C) Moves from side to side
D) Opens only at engine starting

Correct Answer: D (Opens only at engine starting)

Explanation: The air starting valve in a diesel engine is designed to operate intermittently. It opens only during the starting sequence to admit compressed air into the cylinders, initiating rotation; it remains closed during normal operation.

Q6. The purpose of Indicator valve on a generator engine is for _________

A) Fresh air intake
B) Compressed air
C) Blowing through
D) Fuel inlet

Correct Answer: C (Blowing through)

Explanation: The indicator valve on a diesel engine serves as a port to connect a pressure indicator or to release compressed air during the 'blowing through' phase before starting. This allows for clearing residual gases or checking cylinder compression.

Q7. Which of these is NOT a moving part of a diesel engine:

A) Crankcase
B) Connecting rod
C) Piston
D) Crankshaft

Correct Answer: A (Crankcase)

Explanation: The crankcase forms the main body of the engine housing but is typically stationary. The connecting rod, piston, and crankshaft are all essential moving components that translate combustion forces into rotational motion.

Q8. Cylinder Head in a Diesel engine ____

A) Oscillates
B) Rotates in place
C) Moves up and down
D) Stays in place

Correct Answer: D (Stays in place)

Explanation: The cylinder head is a fixed component that seals the top of the engine cylinder. It houses the valves and injectors and remains stationary during the engine's operation.

Q9. The purpose of Inlet valve on a generator engine is for ________

A) Safety
B) Fresh air intake
C) Compress air
D) Fuel inlet

Correct Answer: B (Fresh air intake)

Explanation: The purpose of the inlet valve on a generator engine is to allow the intake of the fresh air-fuel mixture into the cylinder for combustion. This process is critical for the engine's power generation cycle.

Q10. The purpose of Exhaust valve on a generator engine is for _________

A) Combustion Gases
B) Stop fuel
C) Fuel inlet
D) Fresh air intake

Correct Answer: A (Combustion Gases)

Explanation: The exhaust valve on a generator engine opens to allow the expulsion of burnt combustion gases from the cylinder. This clears the cylinder, preparing it for the next intake stroke.

Q11. Before starting a generator engine, LO hand pump should be operated to

A) Warm up the engine
B) Supply lub oil to moving parts
C) Check the level of the tank
D) Turn the engine

Correct Answer: B (Supply lub oil to moving parts)

Explanation: Operating the LO hand pump before starting a generator engine ensures that lubrication oil is distributed to critical moving parts. This pre-lubrication is vital to minimize wear and prevent damage during the initial start-up phase.

Q12. Connecting rod in a Diesel engine ___________.

A) Moves up and down
B) Stays in place
C) Oscillates
D) Rotates in place

Correct Answer: C (Oscillates)

Explanation: The connecting rod in a diesel engine undergoes an oscillating motion. It links the reciprocating piston to the rotating crankshaft, converting the linear movement into rotational movement.

Q13. The purpose of Fuel Oil in a Diesel engine is to _____

A) Reduces friction
B) Gives power
C) Turns turbo charger
D) Causes explosion

Correct Answer: B (Gives power)

Explanation: The primary function of fuel oil in a diesel engine is to provide the energy for combustion. When ignited under pressure, it releases chemical energy, which is converted into mechanical power to drive the engine.

Q14. The purpose of Lubricating Oil in a Diesel engine is to _______

A) Increases air supply
B) Reduces friction
C) Turns turbo charger
D) Gives power

Correct Answer: B (Reduces friction)

Explanation: Lubricating oil plays a crucial role in a diesel engine by reducing friction between moving parts. This minimizes wear, dissipates heat, and prevents seizing, thereby ensuring efficient and prolonged engine operation.

Q15. Crankshaft in a Diesel engine _______

A) Rotates in place
B) Oscillates
C) Moves up and down
D) Stays in place

Correct Answer: A (Rotates in place)

Explanation: The crankshaft in a diesel engine is designed to rotate continuously. It converts the reciprocating motion of the pistons, transmitted through the connecting rods, into the rotational force needed to drive the propeller shaft.

Q16. The purpose of "COMPRESSED AIR" in a diesel engine is to_________

A) Turns turbo Charger
B) Starts Engine
C) Gives Power
D) Controls fuel Supply

Correct Answer: B (Starts Engine)

Explanation: Compressed air is used in diesel engines primarily for starting the engine. It is injected into the cylinders at high pressure to initiate the crankshaft's rotation, allowing the combustion cycle to begin.

Q17. Generator engine on board ships is started using

A) Compressed air
B) Air compressor
C) Hydraulic Oil Motors / Pressure
D) Turbo generator

Correct Answer: A (Compressed air)

Explanation: Generator engines on ships are typically started using a compressed air system. A dedicated starting air receiver stores air under pressure, which is then admitted to the cylinders to turn the engine over.

Q18. Trunk type engines have Piston rod between Piston and Connecting Rod

A) False
B) True

Correct Answer: A (False)

Explanation: Trunk type diesel engines do not have a separate piston rod. The connecting rod is directly attached to the piston skirt, which guides the piston within the cylinder liner.

Q19. Cross head type engines doesn't have Piston Rod between Piston and Connecting Rod

A) False
B) True

Correct Answer: A (False)

Explanation: Crosshead type diesel engines utilize a piston rod which connects the piston to the crosshead. The crosshead then connects to the connecting rod, providing a more robust design for larger engines by separating the piston from the crank mechanism.

Q20. Function of crankcase lubricating oil in a diesel engine is to reduce friction between moving parts, cooling the parts and to _______________

A) Prevent corrosion
B) Supply power
C) Turn the crankshaft
D) Improve combustion

Correct Answer: A (Prevent corrosion)

Explanation: Besides reducing friction and cooling, lubricating oil in the crankcase also serves to prevent corrosion on internal engine components. It forms a protective film that inhibits rust and oxidation, especially during periods of inactivity.

Q21. The purpose of Turbocharger in a Diesel engine is to _________

A) Reduces friction
B) Increase air supply
C) Gives power
D) Turns turbo charger

Correct Answer: B (Increase air supply)

Explanation: The primary function of a turbocharger in a diesel engine is to increase the volume of air supplied to the cylinders. By using exhaust gas energy to drive a compressor, it forces more air into the combustion chamber, enhancing power output and efficiency.

Q22. Turbo Alternator is a generator operating on steam produced from waste heat of Main engine exhaust gases

A) False
B) True

Correct Answer: B (True)

Explanation: A turbo-alternator, also known as an exhaust gas turbo-generator (EGTG), generates electricity by utilizing the waste heat energy from the main engine's exhaust gases to produce steam, which then drives a turbine coupled to an alternator.

Q23. Why diesel engine generator does not have a reversing mechanism

A) Running reverse its phase sequence will change
B) Running reverse it produces DC instead of AC
C) Running reverse it consumes power instead of producing power
D) Running reverse does not produce power

Correct Answer: C (Running reverse it consumes power instead of producing power)

Explanation: Diesel engine generators are not designed with reversing mechanisms because they are coupled to alternators which produce AC power. Running a generator in reverse would cause it to consume electrical power rather than produce it, and it would likely disrupt the ship's electrical grid synchronization.

Q24. Which system is not required for main engine

A) Feed system
B) Lubricating system
C) Cooling system
D) Fuel system

Correct Answer: A (Feed system)

Explanation: A feed system is primarily associated with steam boilers, providing water for steam generation. Main engines on a ship, whether diesel or turbine, do not require a boiler feed system for their direct operation.

Q25. Steam turbine driven generator is called

A) Shaft generator
B) Turbo alternator
C) Generator
D) Alternator

Correct Answer: B (Turbo alternator)

Explanation: A turbo alternator is specifically defined as a generator driven by a steam turbine, which is typically powered by steam produced from the exhaust gas economizer or a fired boiler.

Q26. Main engine is part of __________ system

A) feed water
B) Electricity generation
C) Bilge
D) Propulsion

Correct Answer: D (Propulsion)

Explanation: The main engine is the prime mover dedicated to generating thrust, making it the core component of the ship's propulsion system.

Practice Set 4

Q1. What Machine Tool will you be Using for Making Axial hole in a Round bar.

A) Bending machine
B) Grinding machine
C) Drill machine
D) Lathe machine

Correct Answer: D (Lathe machine)

Explanation: A lathe machine is used for axial drilling, boring, and turning operations on round bars. It precisely rotates the workpiece while a cutting tool moves axially or radially to create the desired hole or shape, ensuring accuracy and a smooth finish.

Q2. What Machine Tool will you be Using for Cutting a joint.

A) Lathe machine
B) Bending machine
C) Drill machine
D) Scissors

Correct Answer: D (Scissors)

Explanation: Scissors are not a machine tool; however, if referring to a machine for cutting, a power saw (like a bandsaw or hacksaw) or a shearing machine would be used for cutting joints in materials. The provided option 'Scissors' is incorrect in a workshop context.

Q3. What machine will you use for cutting a large plate

A) Welding Machine
B) Drill Machine
C) Gas Cutting Machine
D) Grinding Machine

Correct Answer: C (Gas Cutting Machine)

Explanation: A gas cutting machine, such as an oxy-acetylene cutter, is specifically designed for thermally cutting thick metal plates by using a high-pressure oxygen jet. This method is efficient for severing large sections of steel, unlike drilling or grinding.

Q4. What Machine Tool will you be Using for Joining two plates.

A) Drill machine
B) Lathe machine
C) Grinding machine
D) Welding machine

Correct Answer: D (Welding machine)

Explanation: A welding machine is the primary tool used to join two metal plates by melting the edges and fusing them together, often with the addition of filler material. This creates a strong, permanent bond essential for structural integrity.

Q5. What Machine Tool will you be Using for Making holes in a plate

A) Grinding machine
B) Drill machine
C) Bending machine
D) Lathe machine

Correct Answer: B (Drill machine)

Explanation: A drill machine is used to create holes in a plate or other materials. It utilizes a rotating cutting tool (drill bit) to remove material, allowing for the insertion of fasteners or for creating passages.

Q6. To make a new flange for a pipe you will need:

A) Chisel, hammer and drilling machine
B) Lathe machine and drilling machine
C) Grinding machine, drilling machine and pipe cutter
D) Welding machine and milling machine

Correct Answer: B (Lathe machine and drilling machine)

Explanation: To fabricate a new flange for a pipe, a lathe machine is essential for shaping the basic cylindrical form and creating precise mounting surfaces. A drilling machine is then used to accurately create the bolt holes required for connection.

Q7. Which of these is NOT a part of a lathe

A) Anvil
B) Head stock
C) Tool post
D) Chuck

Correct Answer: A (Anvil)

Explanation: An anvil is a heavy metal block used as a surface for forging and hammering metal, and is typically found in a blacksmith's workshop, not a lathe. The headstock, tool post, and chuck are integral parts of a lathe machine's operation.

Q8. What Machine Tool will you be Using for Removing extra material after welding

A) Drill machine
B) Lathe machine
C) Grinding machine
D) Bending machine

Correct Answer: C (Grinding machine)

Explanation: A grinding machine is used for finishing surfaces and removing excess material, particularly after welding. It employs abrasive wheels to precisely remove weld spatter, smooth out rough joints, or achieve a desired surface finish.

Q9. During lathe work, goggles protect our eyes from _______

A) Fumes
B) Heat
C) Metal chips
D) Flash Light

Correct Answer: C (Metal chips)

Explanation: During lathe operations, metal is removed in the form of chips or swarf. Goggles are essential personal protective equipment (PPE) to shield the eyes from these flying metal fragments, preventing serious injury.

Q10. Danger for lifting heavy equipment manually is _________

A) slip
B) fall
C) sprain
D) fracture

Correct Answer: C (sprain)

Explanation: Manually lifting heavy equipment can lead to musculoskeletal injuries such as sprains and strains due to improper lifting techniques or exceeding physical capacity. While fractures are possible with severe accidents, sprains are more common direct consequences of manual lifting.

Q11. DO'S on Machine safety...Mark the incorrect one

A) Keep machine clean and in good condition
B) Check wheel rotation after running of machine
C) Switch off machine immediately if anything goes wrong
D) Check oil levels of machine before starting

Correct Answer: B (Check wheel rotation after running of machine)

Explanation: Checking wheel rotation after a machine has been running is not a standard safety 'do'. It implies touching a potentially moving or hot wheel. All other options are valid safety precautions to ensure the machine operates correctly and safely.

Q12. DONT'S in the workshop...Mark the incorrect one

A) Do not throw things or play practical jokes
B) Do leave your machine unattended
C) Do not run in the workshop
D) Do not touch any equipment unless authorised to do so

Correct Answer: B (Do leave your machine unattended)

Explanation: Leaving a machine unattended while it is running is a significant safety hazard and is strictly prohibited in any workshop environment. It increases the risk of accidents, damage, or unauthorized operation.

Q13. While working with machines do not wash your hands with coolant

A) True
B) False

Correct Answer: A (True)

Explanation: Washing hands with coolant while working with machines is a bad practice. Coolants can contain chemicals that may irritate skin, and they can also make hands slippery, increasing the risk of accidents when operating machinery.

Q14. What machine will you use for cutting a large plate?

A) Welding Machine
B) Drill Machine
C) Gas Cutting Machine
D) Grinding Machine

Correct Answer: C (Gas Cutting Machine)

Explanation: A gas cutting machine, such as an oxy-acetylene cutter, is specifically designed for thermally cutting thick metal plates by using a high-pressure oxygen jet. This method is efficient for severing large sections of steel, unlike drilling or grinding.

Q15. What Machine Tool will you be Using for Joining two plates

A) Drill machine
B) Lathe machine
C) Grinding machine
D) Welding machine

Correct Answer: D (Welding machine)

Explanation: Welding machines provide the necessary heat and filler material to permanently join two plates, which is essential for structural or repair tasks on board.

Q16. When starting to weld you must hold the electrode at an angle of about?

A) 50 degrees
B) 70 degrees
C) 30 degrees
D) 90 degrees

Correct Answer: B (70 degrees)

Explanation: An electrode angle of approximately 70 degrees relative to the direction of travel is standard practice to ensure good arc control, weld penetration, and slag management.

Q17. Oxygen and ______ is used in gas welding on board

A) Argon
B) Acetylene
C) Nitrogen
D) LPG

Correct Answer: B (Acetylene)

Explanation: Oxy-acetylene welding is the standard gas welding method used on ships. Acetylene provides the high-temperature flame necessary to melt steel effectively when mixed with oxygen.

Q18. What Machine Tool will you be Using for Removing extra material after welding.

A) Drill machine
B) Lathe machine
C) Grinding machine
D) Bending machine

Correct Answer: C (Grinding machine)

Explanation: Grinding machines are used to remove weld spatter, slag, and excess weld metal to achieve a smooth finish and inspect the weld profile.

Q19. Before starting arc welding you must:

A) Extend the welding cables
B) Connect the electrode holder to the earth clamp
C) Connect the welding cable to the earth clamp
D) Connect the earth Clamp to the work piece

Correct Answer: D (Connect the earth Clamp to the work piece)

Explanation: For a safe arc, the earth clamp must be securely connected directly to the workpiece to ensure the electrical circuit is complete and to prevent stray current arcs.

Q20. Gases used in gas welding are

A) Acetylene and oxygen
B) CO2 and LPG
C) Oxygen and LPG
D) Acetylene and LPG

Correct Answer: A (Acetylene and oxygen)

Explanation: Gas welding relies on the combustion of oxygen and acetylene, which produces a flame with a temperature high enough to weld steel parts effectively.

Q21. The welding arc length should be:

A) Equal to the diameter of the electrode
B) Less than the diameter of the electrode
C) 20mm
D) 6-12mm

Correct Answer: A (Equal to the diameter of the electrode)

Explanation: For stable arc welding, the arc length should be maintained roughly equal to the diameter of the electrode core wire to prevent arc instability or poor fusion.

Q22. The function of Electrode dryer is to wet the electrodes that have been exposed to a humid atmosphere

A) False
B) True

Correct Answer: A (False)

Explanation: The purpose of an electrode dryer is the exact opposite: it is used to remove moisture from electrodes to prevent hydrogen-induced cracking, which occurs if electrodes are damp.

Q23. Too long an arc will cause

A) This is bulb up the weld pool
B) Low penetration coarse splatter & flat bead
C) Uncontrolled penetration slag globules electrode stuck to weld
D) Even bead & good penetration & good welding

Correct Answer: B (Low penetration coarse splatter & flat bead)

Explanation: An excessively long arc results in increased voltage, leading to low penetration, unstable arc characteristics, and significant coarse splatter. The instability often produces a flat, uneven weld bead because the heat is not sufficiently concentrated to fuse the base metal effectively.

Q24. Too short an arc length will cause an uneven bead due to arc failure

A) None of the above
B) Good welding
C) Low penetration coarse splatter
D) Flat bead

Correct Answer: A (None of the above)

Explanation: A short arc length is preferred for stability and control. The provided options for this specific question imply that none of the listed consequences (low penetration or flat bead) are inherent to a short arc, as a short arc generally provides better control and deeper penetration if maintained correctly.

Q25. Electrode should be kept at a distance of ________ above the spot where we wish to strike the arc

A) 30-40 mm
B) 20-30 mm
C) 40-50 mm
D) 35-45 mm

Correct Answer: B (20-30 mm)

Explanation: When preparing to strike an arc, the electrode tip should be positioned approximately 20-30 mm above the intended starting point. This distance allows the welder to stabilize their position and prepare for the initial strike while minimizing the risk of accidental arcing or sticking.

Q26. Welding gloves made from tanned leather can be following combinations

A) 3-finger and 5-finger type
B) 2-finger and 4-finger type
C) 4-finger and 6-finger type
D) 1-finger and 2-finger type

Correct Answer: A (3-finger and 5-finger type)

Explanation: Welding gloves used in manual metal arc welding are designed for dexterity and protection, commonly available in 3-finger and 5-finger configurations. These provide the necessary heat resistance and allow for the specific grip required for electrode holders.

Q27. Spats protect the ankles from melted metal and splatter

A) True
B) False

Correct Answer: A (True)

Explanation: Spats are essential items of Personal Protective Equipment (PPE) for welders. They are worn over safety boots to protect the ankles and the top of the footwear from molten metal droplets and slag splatter, preventing burn injuries.

Q28. In welding screen, welding glasses are always used with Protective glasses

A) False
B) True

Correct Answer: A (False)

Explanation: Welding screens are fitted with high-density filter lenses (shades) that provide protection against harmful ultraviolet and infrared radiation. In standard maritime safety practice, the welding lens itself acts as the primary protection, and additional clear protective glasses are not required underneath.

Practice Set 5

Q1. The level measuring device, Gauge glass & can be used for

A) Tanks of large depth
B) Multi Purpose
C) Tanks of Small depths
D) Oil can

Correct Answer: C (Tanks of Small depths)

Explanation: Gauge glasses utilize the principle of communicating vessels to indicate liquid levels. Due to their physical size and the potential for glass breakage under hydrostatic pressure, they are generally restricted to small, shallow tanks or auxiliary equipment.

Q2. The level measuring device, Level Indicator can be used for

A) Oil level in generator crankcase
B) Sumps of purifiers, compressors, etc
C) Oil can
D) Small to large tanks within the engine room

Correct Answer: B (Sumps of purifiers, compressors, etc)

Explanation: Level indicators in this context are typically sight glasses or float-type local indicators used for small, localized machinery sumps. These allow for quick visual checks of oil levels in equipment like purifiers and compressors.

Q3. The level measuring device, Sounding tape can be used for

A) Oil can
B) Tanks of large depth
C) Multi Purpose
D) Oil level in generator crankcase

Correct Answer: B (Tanks of large depth)

Explanation: A sounding tape is a flexible, calibrated steel tape with a weighted plummet. It is the industry standard for measuring liquid depths in large-capacity storage tanks, such as double-bottom or deep tanks, where accuracy over a long vertical distance is required.

Q4. The level measuring device, Remote level indicators can be used for

A) Multi Purpose
B) Oil can
C) Tanks of large depth
D) Oil level in generator crankcase

Correct Answer: A (Multi Purpose)

Explanation: Remote level indicators utilize various technologies (ultrasonic, hydrostatic, or radar) to transmit data to a central control station. Their versatility allows them to be used across a multi-purpose range of tank types and depths throughout the ship.

Q5. A sounding tape is used for measuring level of a liquid in a:

A) Tank
B) Boiler
C) Crank case
D) Cylinder

Correct Answer: A (Tank)

Explanation: A sounding tape is a specialized measuring tool consisting of a graduated tape and a heavy bob. It is used specifically to measure the liquid level or ullage in ships' tanks to ensure compliance with loading and stability requirements.

Q6. A gauge-glass for checking liquid level is fitted on which tanks

A) Fore Peak Ballast Tank
B) FO Storage Double Bottom Tank
C) After Peak Ballast Tank
D) Lub Oil Storage Tank

Correct Answer: D (Lub Oil Storage Tank)

Explanation: Gauge glasses are frequently installed on lubricating oil storage tanks in the engine room for convenient visual monitoring. Ballast and fuel tanks typically use sounding pipes or remote sensors to avoid the risk of oil or water leakage associated with glass tubes.

Q7. A dip stick is used for checking level of oil in a

A) Filter
B) Tank
C) Cylinder
D) Crank case

Correct Answer: D (Crank case)

Explanation: A dipstick is a simple rod used to check the oil level directly within a machinery crankcase. It works by showing the oil level against marked graduation lines, providing a quick and reliable reading for reciprocating machinery.

Q8. The level measuring device, Dip stick can be used for

A) Multi Purpose
B) Oil level in generator crankcase
C) Tanks of large depth
D) Oil can

Correct Answer: B (Oil level in generator crankcase)

Explanation: Dipsticks are specifically designed for small volumes and enclosed spaces like machinery crankcases. They provide a direct, tactile method for checking oil levels to ensure proper lubrication before and during machinery operation.

Q9. The level measuring device, Sounding rod can be used for

A) Oil can
B) Tanks of large depth
C) Oil level in generator crankcase
D) Multi Purpose

Correct Answer: D (Multi Purpose)

Explanation: Sounding rods are versatile tools used to measure the depth of liquids in various tanks, including deep tanks and double-bottom tanks. Their simplicity and robustness make them suitable for multiple applications throughout the ship's tank systems.

Q10. The most common item provided for checking oil level in a small machine is

A) Sounding tape
B) Sounding rod
C) Sight glass
D) Remote level indicator

Correct Answer: C (Sight glass)

Explanation: A sight glass is a transparent tube or window attached to the side of a small machine or reservoir. It allows the operator to see the oil level instantly without opening the system, making it the most common and convenient indicator for small equipment.

Q11. What material is not a commonly used as insulation material?

A) Plastic
B) Wood
C) Glass Wool
D) Fire Bricks

Correct Answer: A (Plastic)

Explanation: Plastic is a poor insulation material for high-temperature maritime applications and can melt or release toxic fumes if exposed to heat. Materials like glass wool and fire bricks are standard for their thermal resistance and safety properties.

Q12. What is used to prevent the transfer of heat from pipes or equipment?

A) Insulating material
B) Cooling system
C) Turbo charger
D) Air circulation

Correct Answer: A (Insulating material)

Explanation: Insulating materials are substances with low thermal conductivity used to reduce the rate of heat transfer. They are essential on ships to keep heat within pipes and equipment, thereby improving thermal efficiency and preventing personnel burns.

Q13. What is the common name for item used for preventing the loss of heat from pipes or equipment?

A) Lagging
B) Cooling system
C) Exhaust boiler
D) Air circulation

Correct Answer: A (Lagging)

Explanation: Lagging is the maritime term for the thermal insulation applied to pipes, boilers, and machinery. It serves the primary purpose of heat conservation and personnel protection by reducing surface temperatures to safe limits.

Q14. What materials are banned for use as insulation materials?

A) Fire Bricks
B) Thermocole
C) Fiberglass Wool
D) Asbestos

Correct Answer: D (Asbestos)

Explanation: Asbestos is a hazardous material known to cause severe respiratory illnesses and cancer. It is strictly prohibited under SOLAS regulations in the construction and maintenance of modern ships; non-hazardous alternatives must be used.

Q15. Woolen garments, blankets and quilts are termed as Insulating Materials

A) True
B) False

Correct Answer: A (True)

Explanation: Wool is a natural fiber with excellent insulating properties due to its ability to trap air. When used as clothing, blankets, or quilts, it acts as a thermal barrier, effectively reducing heat transfer.

Q16. While working with Insulating material such as glass wool, you must not protect your body using gloves and goggles

A) True
B) False

Correct Answer: B (False)

Explanation: Glass wool contains fine, sharp fibers that cause skin irritation, respiratory distress, and eye damage. It is mandatory to wear appropriate PPE, including gloves, goggles, and respiratory protection, whenever handling this insulating material.

Q17. Lagging is the Insulating material wrapped around pipes, boilers or tanks to allow loss of heat

A) True
B) False

Correct Answer: B (False)

Explanation: The purpose of lagging is to prevent, not allow, the loss of heat. By covering equipment with insulating material, engineers maintain thermal efficiency and protect the crew from high-temperature surfaces.

Q18. Electrical Insulation prevent flow of ......

A) Power
B) Resistance
C) Electricity
D) Voltage

Correct Answer: C (Electricity)

Explanation: Electrical insulation refers to materials used to wrap conductors to prevent the flow of electricity to unwanted paths. This prevents short circuits, equipment damage, and electrical shocks to personnel.

Q19. While assembling pipes, take care in removing the lagging and putting it back after completing the work

A) True
B) False

Correct Answer: B (False)

Explanation: Proper maintenance protocol dictates that lagging removed during pipe repairs must be re-fitted or replaced. Failure to do so leads to excessive heat loss and creates a safety hazard due to exposed high-temperature pipework.

Q20. Insulating material used is generally Fire resistant

A) True
B) False

Correct Answer: A (True)

Explanation: Insulating materials used in engine rooms must be fire-resistant or non-combustible to prevent the spread of fire. This is a critical requirement under FSS Code and international fire safety standards for ships.

Q21. Chemicals should be stored in _______

A) in accommodation
B) in engine room
C) in separate well ventilated store
D) on deck

Correct Answer: C (in separate well ventilated store)

Explanation: Chemicals must be stored in a dedicated, well-ventilated storeroom to prevent the accumulation of hazardous vapors and to ensure safety. Storing them in accommodation or the engine room poses significant health and fire risks.

Q22. If any chemical fall on the body, first action should be

A) apply medicine to affected area
B) wipe clean the affected area
C) call for the doctor
D) thoroughly wash affected area

Correct Answer: D (thoroughly wash affected area)

Explanation: In the event of chemical contact, immediate flushing with large quantities of clean, fresh water is the universal first-aid procedure. This dilutes and removes the chemical agent, minimizing potential chemical burns.

Q23. Chemicals on-board are not to be used for

A) cleaning the boiler
B) cleaning the cooking utensils
C) boiler water treatment
D) fuel oil treatment

Correct Answer: B (cleaning the cooking utensils)

Explanation: Chemicals used for industrial applications, such as boiler water treatment or fuel oil treatment, are often toxic and hazardous. They must never be used for any domestic purposes, such as cleaning cooking utensils, due to the risk of poisoning.

Q24. Disposal of used/expired chemical container should be done by

A) throwin into deep sea
B) disposal with ships garbage
C) disposal as per makers instruction
D) burning in incinerator

Correct Answer: C (disposal as per makers instruction)

Explanation: Used or expired chemical containers must be disposed of strictly according to the manufacturer's Safety Data Sheet (SDS) and the ship's Garbage Management Plan. This ensures environmental compliance and avoids safety hazards.

Q25. Chemicals are mainly used for

A) Fuel treatment
B) Boilers
C) Testing quality of water
D) All of the above

Correct Answer: D (All of the above)

Explanation: Chemicals on board are essential for maintaining the integrity of mechanical systems. This includes fuel oil treatment, boiler water quality monitoring, and cooling water maintenance, all of which are crucial for reliable ship operation.

Q26. Use of personal protective gear while handling chemicals...Mark the incorrect one

A) Face mask
B) Apron
C) Rubber gloves
D) Sun protect Goggles

Correct Answer: D (Sun protect Goggles)

Explanation: Sun protection goggles are intended for outdoor UV protection and provide no defense against chemical splashes. Proper PPE for chemical handling includes chemical-resistant goggles, gloves, and aprons to prevent injury.

Q27. Steering gear is used to turn the

A) crane boom/jib
B) main engine
C) propeller
D) rudder

Correct Answer: D (rudder)

Explanation: The steering gear is the machinery responsible for rotating the rudder to control the heading of the ship. It is a critical component for the safe navigation and maneuvering of the vessel.

Q28. Steering gear is located in the

A) forward of engine room
B) on the bridge
C) in a separate compartment aft of engine room
D) forecastle

Correct Answer: C (in a separate compartment aft of engine room)

Explanation: Steering gear is located in the steering gear flat, which is situated at the extreme aft of the ship. This position allows for a direct, mechanical, or hydraulic connection to the rudder stock.

Q29. Rudder of the ship is turned by means of a

A) hydraulic pump
B) propeller
C) Cylinder and Ram arrangement only
D) steering wheel

Correct Answer: C (Cylinder and Ram arrangement only)

Explanation: In most modern ships, the steering gear utilizes a hydraulic system where power units drive hydraulic rams. These rams act on the tiller to move the rudder, translating hydraulic force into mechanical steering motion.

Q30. Which of the following is a correct statement

A) Steering gear is located at aft part of the ship
B) Rudder is connected to the propeller
C) Rudder is located in the steering flat
D) Propeller is turned by the rudder

Correct Answer: A (Steering gear is located at aft part of the ship)

Explanation: The steering gear flat is located at the aft-most part of the ship, directly above or near the rudder assembly. This allows the steering gear hydraulic rams to engage with the rudder stock efficiently.

Q31. Various components of Steering Gear ...Tick the incorrect one

A) Hydrophore Unit
B) Actuating Unit
C) Power Unit
D) Control Unit

Correct Answer: A (Hydrophore Unit)

Explanation: A hydrophore unit is part of the ship's freshwater distribution system, not the steering gear. The steering gear consists of the power unit, control unit, and actuating (or tiller) unit.

Q32. The thruster placed in the aft of the ship is called Bow Thruster

A) False
B) True

Correct Answer: A (False)

Explanation: A thruster placed at the aft of the ship is called a stern thruster. A bow thruster is exclusively located in the forward part of the ship to facilitate docking and maneuvering.

Q33. The thruster placed in the forward end of the ship is called Stern Thruster

A) True
B) False

Correct Answer: B (False)

Explanation: This statement is false because a thruster placed at the forward end of the ship is defined as a bow thruster. A stern thruster is positioned at the aft end of the vessel.

Q34. While taking rounds in the Steering flat, following to be checked...Mark the incorrect

A) Leakages from pumps, pipe joints, etc
B) Load of the pump in Ampere
C) Oil level of MDO in the steering gear system
D) Abnormal noise or vibrations

Correct Answer: C (Oil level of MDO in the steering gear system)

Explanation: Checking the oil level of MDO (Marine Diesel Oil) is irrelevant in the steering gear flat, as steering systems use specific hydraulic oil. Operators should check the hydraulic oil level, not fuel oil levels.

Practice Set 6

Q1. A quick closing valve is fitted on

A) boiler steam line
B) main engine cooling system
C) fuel tanks above double bottom tanks
D) compressed air line

Correct Answer: C (fuel tanks above double bottom tanks)

Explanation: Quick closing valves are mandatory on fuel tanks located above double-bottom tanks. They provide an immediate, remote means to shut off fuel supply in the event of an engine room fire.

Q2. Man hole door on a tank is fitted for

A) allowing entry to the tank when needed
B) preventing entry by unauthorized persons
C) preventing damage on tanks
D) strengthening the tank

Correct Answer: A (allowing entry to the tank when needed)

Explanation: A manhole door provides access to the interior of a tank for essential maintenance, such as cleaning, inspection, or repairs. These doors must be pressure-tight when closed to ensure tank integrity.

Q3. A quick closing valve

A) prevents fire from occuring
B) can be remotely closed in an emergency
C) prevents oil spills in engine room
D) stops air supply to main engine

Correct Answer: B (can be remotely closed in an emergency)

Explanation: Quick closing valves are essential safety devices designed to be shut remotely from outside the engine room. They are triggered during fire emergencies to cut off fuel supply to machinery and prevent the escalation of a fire.

Q4. Air vent to tanks is provided for

A) preventing build up of pressure or vaccum
B) allowing liquid to come out when tank is full
C) preventing fire
D) for ventilation

Correct Answer: A (preventing build up of pressure or vaccum)

Explanation: Air vent pipes are installed on tanks to allow air to escape when filling and to let air enter when emptying. This prevents dangerous pressure buildup or vacuum formation which could cause tank structural damage.

Q5. Sounding pipes on tanks are fitted for finding

A) level of liquid in the tank
B) pressure in the tank
C) temperature in the tank
D) abnormal sound in the tank

Correct Answer: A (level of liquid in the tank)

Explanation: Sounding pipes provide a direct, reliable method for measuring the depth of liquid in a tank. By inserting a sounding tape into the pipe, the crew can determine the quantity of fuel or water present.

Q6. Quick closing valve operation is carried out from

A) remote location outside engine room
B) anywhere in the engine room
C) steering flat
D) navigation bridge

Correct Answer: A (remote location outside engine room)

Explanation: For safety, quick closing valves must be operable from a location outside the machinery space. This ensures that in a fire scenario, the fuel supply can be shut down without endangering personnel.

Q7. Ballast tank on a ship are filled up with _____

A) drinking water
B) sea water
C) cargo oil
D) water from other tanks

Correct Answer: B (sea water)

Explanation: Ballast tanks are filled with sea water to maintain the ship's stability, trim, and draft requirements. This allows the ship to operate safely even when not carrying a full cargo load.

Q8. Settling tanks on ship are provided for

A) allowing water and sludge to separate before oil seperator
B) mixing oils before use
C) storing oils before its use
D) filtering water from fuel oil

Correct Answer: A (allowing water and sludge to separate before oil seperator)

Explanation: Settling tanks provide the residence time necessary for water and sludge to settle at the bottom of the fuel. This separation process ensures that the fuel oil purifier receives cleaner oil for final treatment.

Q9. While bunkering is going on, what will be the most important thing to monitor

A) Oil sample is being taken
B) Temperature of oil being supplied
C) Oil sounding in the tank being filled
D) Oil sounding in bunker barge

Correct Answer: C (Oil sounding in the tank being filled)

Explanation: Monitoring the tank sounding is critical during bunkering to prevent overflows. Preventing spills is a high priority for MARPOL compliance and shipboard safety, requiring constant vigilance of tank levels.

Q10. Quick Closing Valves can be operated...

A) All of the above
B) Locally
C) Pneumatically
D) Remotely

Correct Answer: A (All of the above)

Explanation: Quick closing valves are versatile safety devices designed for emergency operations. They can be closed locally by hand or remotely via pneumatic or hydraulic systems to ensure maximum safety in diverse fire scenarios.

Q11. Each tank is required to have at least the following

A) All of the above
B) Outlet Line
C) Air Vent
D) Sounding Pipe

Correct Answer: A (All of the above)

Explanation: Each tank requires an outlet line for discharge, an air vent to prevent over-pressurization during filling, and a sounding pipe to manually check levels. As all these components are fundamental for safe tank operation, 'All of the above' is the correct answer.

Q12. Storage Tank in the engine room will be called as ______ if the tank capacity is in between 2-50 tonnes

A) Ballast Tank
B) Medium Tank
C) Large Tank
D) Small Tank

Correct Answer: D (Small Tank)

Explanation: Under standard maritime nomenclature for engine room storage containers, tanks with a capacity ranging between 2 and 50 tonnes are classified as Small Tanks.

Q13. On finding an injured person in the engine room, your first action will be to

A) apply first aid
B) report to the duty officer on watch
C) bring a stretcher
D) look for the cause of injury

Correct Answer: B (report to the duty officer on watch)

Explanation: Safety procedures prioritize immediate communication. Reporting to the duty officer ensures help is mobilized quickly and proper emergency procedures are activated without delay, which is the primary duty before attempting independent medical aid.

Q14. A fire alarm in the engineroom is activated

A) from the bridge
B) automatically on machinery failure
C) due to a black out
D) due to fire in the engine room

Correct Answer: D (due to fire in the engine room)

Explanation: A fire alarm in the engine room is specifically triggered by fire detection sensors (heat or smoke) identifying an active fire, requiring immediate emergency response protocols.

Q15. On hearing a CO2 alarm in the engine room, your first action will be

A) report to the duty officer on watch
B) use large fire extinguisher on the fire
C) use emergency exit to get out
D) raise fire alarm

Correct Answer: C (use emergency exit to get out)

Explanation: A CO2 fire extinguishing system releases asphyxiating gas into the engine room space. Therefore, the immediate priority upon hearing the warning alarm is to evacuate via the emergency exit to avoid suffocation.

Q16. A machinery failure alarm in the engine room is activated

A) from the bridge
B) automatically on machinery failure
C) due to collision of the ship
D) due to fire in the engine room

Correct Answer: B (automatically on machinery failure)

Explanation: Machinery failure alarms are designed to alert engineers to faults in running equipment (e.g., low lube oil pressure or high jacket water temperature). These are generated automatically by monitoring systems.

Q17. A general alarm is activated

A) from the bridge
B) automatically on machinery failure
C) due to black out
D) due to high bilge level

Correct Answer: A (from the bridge)

Explanation: The General Alarm (GA) is the highest-level emergency signal on a ship. It is manually activated from the bridge by the master or duty officer to alert all personnel of an emergency situation requiring mustering.

Q18. UMS stands for

A) Uniform Machinery Space Operation
B) Unattended Machinery Space Operation
C) Undermanned Machinery Space Operation
D) Under Machinery Space Operation

Correct Answer: B (Unattended Machinery Space Operation)

Explanation: UMS refers to Unattended Machinery Space operations, where the engine room is not manned by engineers on a watch-keeping basis, relying instead on comprehensive alarm and monitoring systems.

Q19. Accident can't happen in the engine room due to falling objects

A) False
B) True

Correct Answer: A (False)

Explanation: Falling objects represent a significant hazard in the engine room, especially during maintenance or heavy weather. Safety helmets and proper stowage of tools are mandatory precisely because accidents of this nature occur.

Q20. Which of the following is NOT an audio visual alram

A) Fire Alarm
B) Co2 Alarm
C) Ship security alarm
D) Machinery Alarm

Correct Answer: C (Ship security alarm)

Explanation: The Ship Security Alarm (SSAS) is a covert, silent alarm system used to alert authorities of piracy or security threats, intentionally excluding audible or visual indicators to protect the crew.

Q21. General Alarm and Co2 alarm are same

A) True
B) False

Correct Answer: B (False)

Explanation: The General Alarm is a seven-short-blast and one-long-blast signal for muster, whereas the CO2 alarm is a continuous or distinct alarm specific to the release of fixed fire-extinguishing gas. They serve different critical safety functions.

Practice Set 7

Q1. Emergencies that can occur in the Engine room

A) Stoppage of Main engine
B) Crank case explosion
C) Flooding of the engine room
D) All of the above

Correct Answer: D (All of the above)

Explanation: Crankcase explosions, main engine failure, and flooding are all critical operational emergencies requiring immediate intervention to maintain ship safety and environmental compliance.

Q2. Fixed fire fighting system for engine room

A) DCP
B) CO2
C) inert gas
D) nitrogen

Correct Answer: B (CO2)

Explanation: Fixed CO2 systems are the industry standard for engine room fire suppression due to the gas's ability to reach inaccessible areas and effectively smother Class B and C fires without leaving residues.

Q3. Not a part of a fire triangle

A) fuel
B) heat
C) electricity
D) air

Correct Answer: C (electricity)

Explanation: The fire triangle consists of fuel, heat, and oxygen. Electricity is an energy source that may ignite fuel but is not an element of the triangle itself.

Q4. Main fire pump is located

A) on main deck
B) at the fire station
C) in the engine room
D) in the steering flat

Correct Answer: C (in the engine room)

Explanation: Under SOLAS regulations, the main fire pumps must be located within the machinery space to be readily available for engine room firefighting, distinct from the emergency fire pump.

Q5. To fight an electric fire you would use

A) dry powder
B) steam jet
C) water spray
D) special foam

Correct Answer: A (dry powder)

Explanation: Dry powder is an electrical non-conductor and is the preferred medium for Class C electrical fires to suppress flames without damaging equipment or risking electrocution to the operator.

Q6. Means of forcing water out of a portable extinguisher

A) CO2 cartridge
B) air cartridge
C) chemical cartridge
D) nitrogen cartridge

Correct Answer: A (CO2 cartridge)

Explanation: Most portable fire extinguishers on board use a pressurized gas cartridge, such as CO2, to expel the contents from the canister onto the fire source.

Q7. High expansion foam system is suitable for fighting fires in

A) accommodation
B) cargo holds
C) engine room
D) deck of ship

Correct Answer: C (engine room)

Explanation: High expansion foam systems are specifically designed to fill large, confined volumes like engine rooms, effectively displacing oxygen and cooling surfaces to extinguish major fires.

Q8. Which of the following provided for connecting to shore water supply

A) foam tank
B) fixed foam system
C) fire hydrant
D) international shore connection

Correct Answer: D (international shore connection)

Explanation: The International Shore Connection (ISC) is a standardized flange required by MARPOL/SOLAS regulations to allow port authorities to connect their shore water supply to the ship's fire main.

Q9. The emergency fire pump on a ship is located

A) at the poop deck
B) inside the engine room
C) outside the engine room
D) at the boat deck

Correct Answer: C (outside the engine room)

Explanation: The emergency fire pump is specifically located outside the engine room to ensure that, in the event of a total engine room fire, a firefighting water supply remains available.

Q10. Your first action upon finding a fire

A) use portable extinguisher
B) use portable extinguisher
C) raise an alarm
D) switch off electrical power supply

Correct Answer: C (raise an alarm)

Explanation: Raising the alarm is the mandatory first action in fire fighting procedures. This ensures that the crew is alerted, the bridge is informed, and the emergency team can be organized before fighting the fire.

Q11. Fire hose on ship are made of

A) strong canvas
B) thick cloth
C) leather
D) flexible teel

Correct Answer: A (strong canvas)

Explanation: Fire hoses used on ships must be flexible and durable; they are constructed from high-strength canvas or synthetic fabric to withstand high pressure and environmental wear.

Q12. What is used in case the fire in the engine room becomes out of control

A) portable fire extinguisher
B) 45 liters foam fire extinguisher
C) fixed CO2 fire extinguishing system
D) fire pumps and hoses

Correct Answer: C (fixed CO2 fire extinguishing system)

Explanation: Fixed CO2 fire extinguishing systems are designed for total flooding of the engine room when a fire is beyond the capacity of portable equipment or localized firefighting systems.

Q13. Which one of the following will be used first on a small fire in the engine room

A) portable fire extinguisher
B) fire blancket
C) fixed fire extinguishing system
D) fire pumps and hoses

Correct Answer: A (portable fire extinguisher)

Explanation: Portable fire extinguishers are the initial line of defense for small, localized fires, allowing for immediate action before a fire can propagate across the engine room.

Q14. Full form of PASS....Mark the correct one

A) Pull, Aim, Smoke,
B) Pull, Aim, Squeeze, Sweep
C) Pull, Aim, Sponge, Sweep
D) Push, Automatic, Sweep, Squeeze

Correct Answer: B (Pull, Aim, Squeeze, Sweep)

Explanation: PASS is the universal acronym for operating a portable fire extinguisher: Pull the pin, Aim at the base of the fire, Squeeze the handle, and Sweep from side to side.

Q15. Following is not a part of Fire detection system

A) Heat Detector
B) Flame Detector
C) Body Detector
D) Smoke Detector

Correct Answer: C (Body Detector)

Explanation: Fire detection systems utilize sensors for heat, smoke, and flame. A 'body detector' is not a component of maritime fire detection equipment.

Q16. Which Instrument is used for testing Insulation

A) Driver
B) Megger
C) Drager
D) Jagger

Correct Answer: B (Megger)

Explanation: A Megger (megohmmeter) is the standard electrical test instrument used to measure the insulation resistance of electrical cables and windings.

Q17. Before starting maintenance on any equipment, following precautions to be taken...Mark the incorrect one

A) Remove the safety guards
B) Put a warning notice on the Mains switch
C) Switch on the main supply
D) Switch off the machine

Correct Answer: C (Switch on the main supply)

Explanation: Switching on the main supply during maintenance is dangerous and violates all LOTO (Lock-Out, Tag-Out) safety protocols, which strictly mandate isolation of power to prevent accidental energization.

Q18. Before re-starting a machine, following precautions to be taken...Mark the incorrect one

A) Check that no tools have been left on or near the machine
B) Replace the safety guard after starting the machine
C) Restore the power supply
D) Remove any warning notices

Correct Answer: B (Replace the safety guard after starting the machine)

Explanation: Safety guards must be reinstalled and secured before starting any machine. Running machinery without guards exposes personnel to rotating parts and creates a severe safety hazard.

Q19. Dry full contact body resistance is about ______ ohms at 250V

A) 4000
B) 5000
C) 2000
D) 3000

Correct Answer: C (2000)

Explanation: The average electrical resistance of the human body when dry is approximately 2000 ohms. This value decreases significantly if the skin is wet, increasing the risk of fatal shock.

Q20. What is the minimum value of shock current (AC) which can be fatal

A) 25 mA

B) 35 mA

C) 20 mA

D) 15mA

Correct Answer: D (15mA)

Explanation: Current as low as 15mA can cause involuntary muscle contraction and interfere with cardiac function, making it a threshold for potentially fatal electric shock.